

R Notebook

Code ▼

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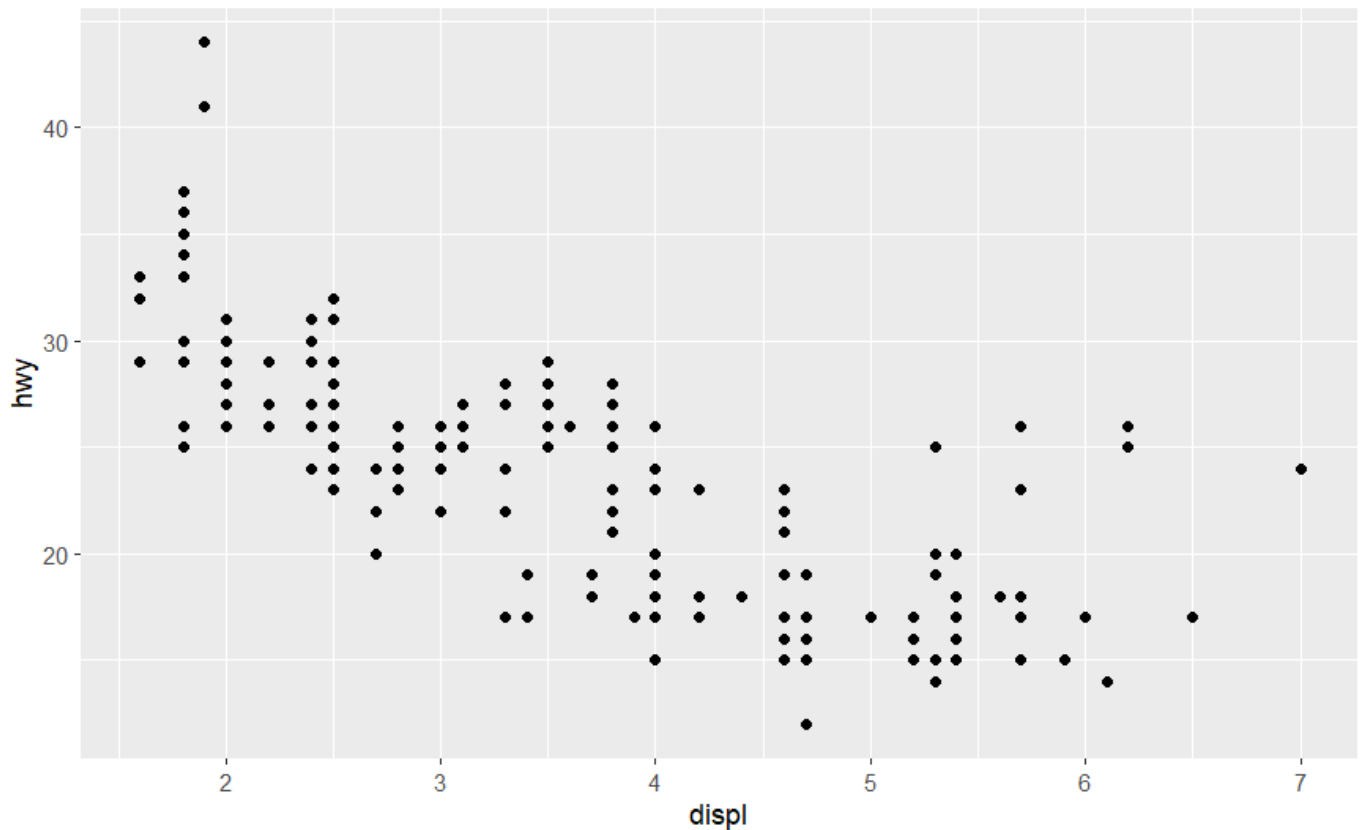
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```
library(tidyverse)
#library(datasets)
str(mpg)
```

```
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ      : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year       : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl        : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans      : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv        : chr [1:234] "f" "f" "f" "f" ...
 $ cty        : int [1:234] 18 21 20 21 16 18 18 16 20 ...
 $ hwy        : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl         : chr [1:234] "p" "p" "p" "p" ...
 $ class      : chr [1:234] "compact" "compact" "compact" "compact" ...
```

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```
#Multiple ways to get the same thing done.
#The long way
ggplot() +
  layer(
    data = mpg, mapping = aes(x = displ, y = hwy),
    geom = "point", stat = "identity", position = "identity"
  ) +
  scale_y_continuous() +
  scale_x_continuous() +
  coord_cartesian()
```

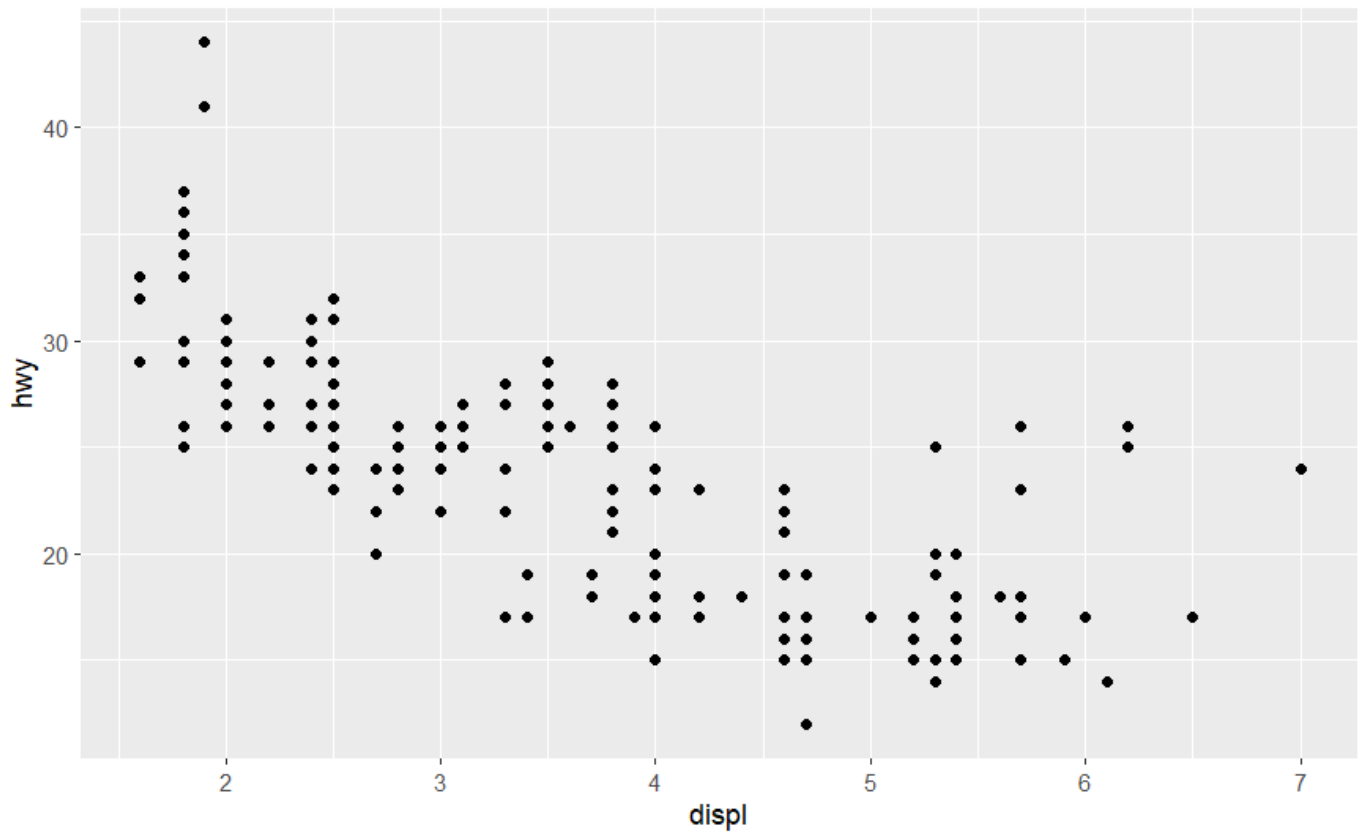


Q1 what are the drawbacks of the code above especially in the situation where we are doing EDA?

The code works, but it's way too long and repetitive — for EDA you want short, simple code so you can quickly try out different plots without extra clutter.

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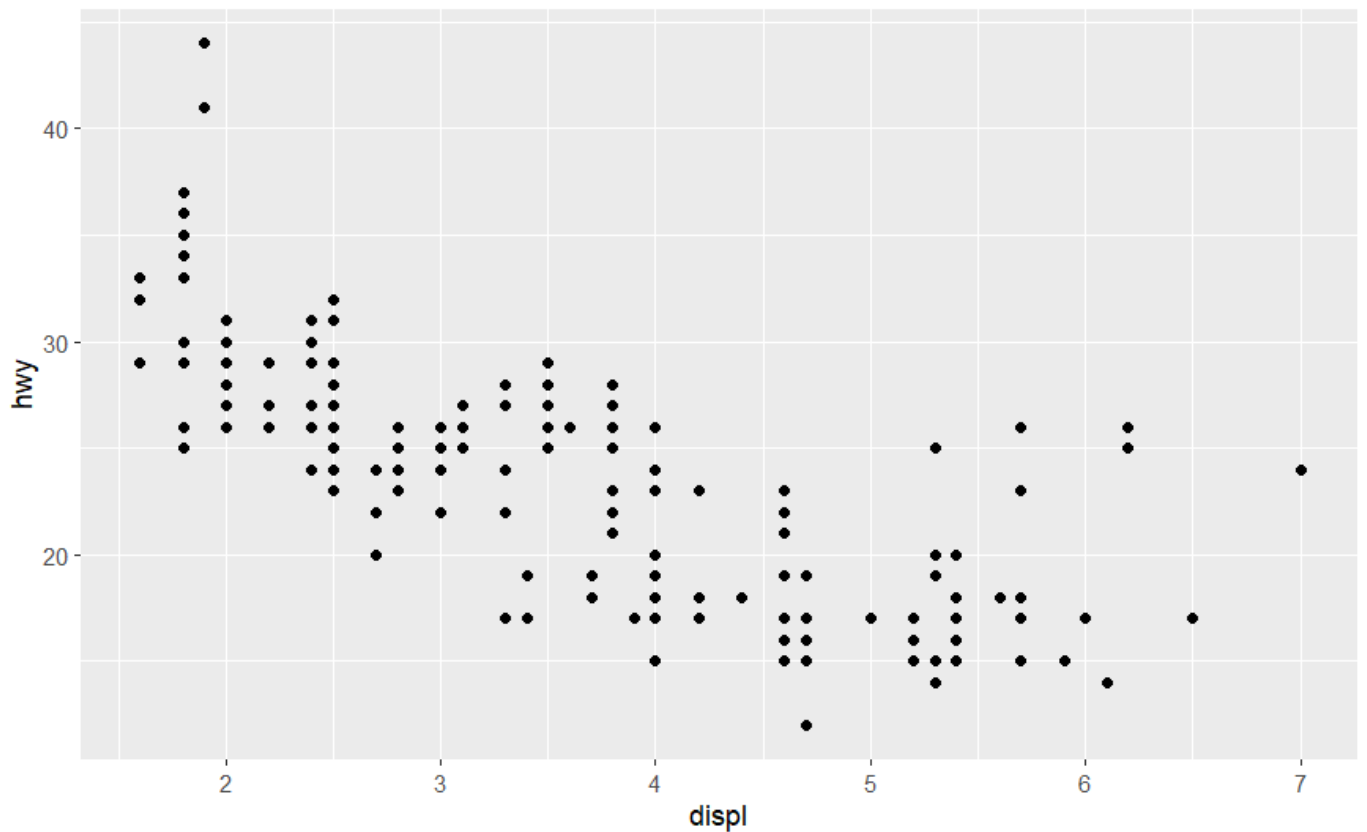
```
# Shorthand 1
g <- ggplot(data = mpg, aes(x=displ, y=hwy))
g + geom_point()
```



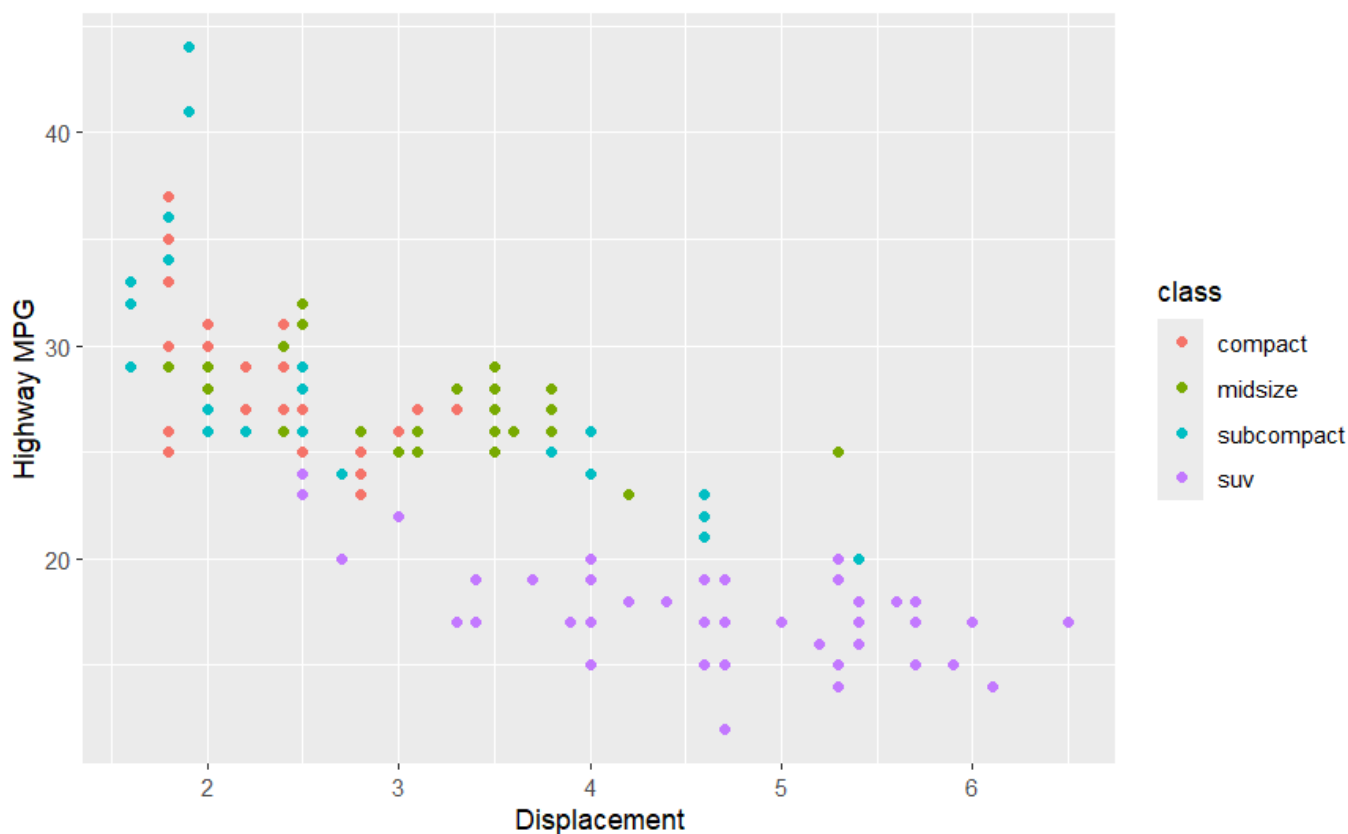
It does the same thing without specifying the layers.

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```
#Shorthand 2
g <- ggplot(data = mpg)
g + geom_point(aes(x=displ, y=hwy))
```

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```
# Let's tweak things around!
# What does class %in% c(...) do?
# What does labs(...) do?
g <- ggplot(data = subset(mpg,
class %in% c("compact",
"midsize", "subcompact",
"suv")))
g + geom_point(aes(x=displ,
y=hwy, color = class)) + labs(x="Displacement", y="Highway MPG")
```



Were Tufte's "rules" met? Specifically his rule about the number of dimensions. → Yes — the plot uses just two dimensions (x and y) plus one grouping variable (color), so it's not overloaded.

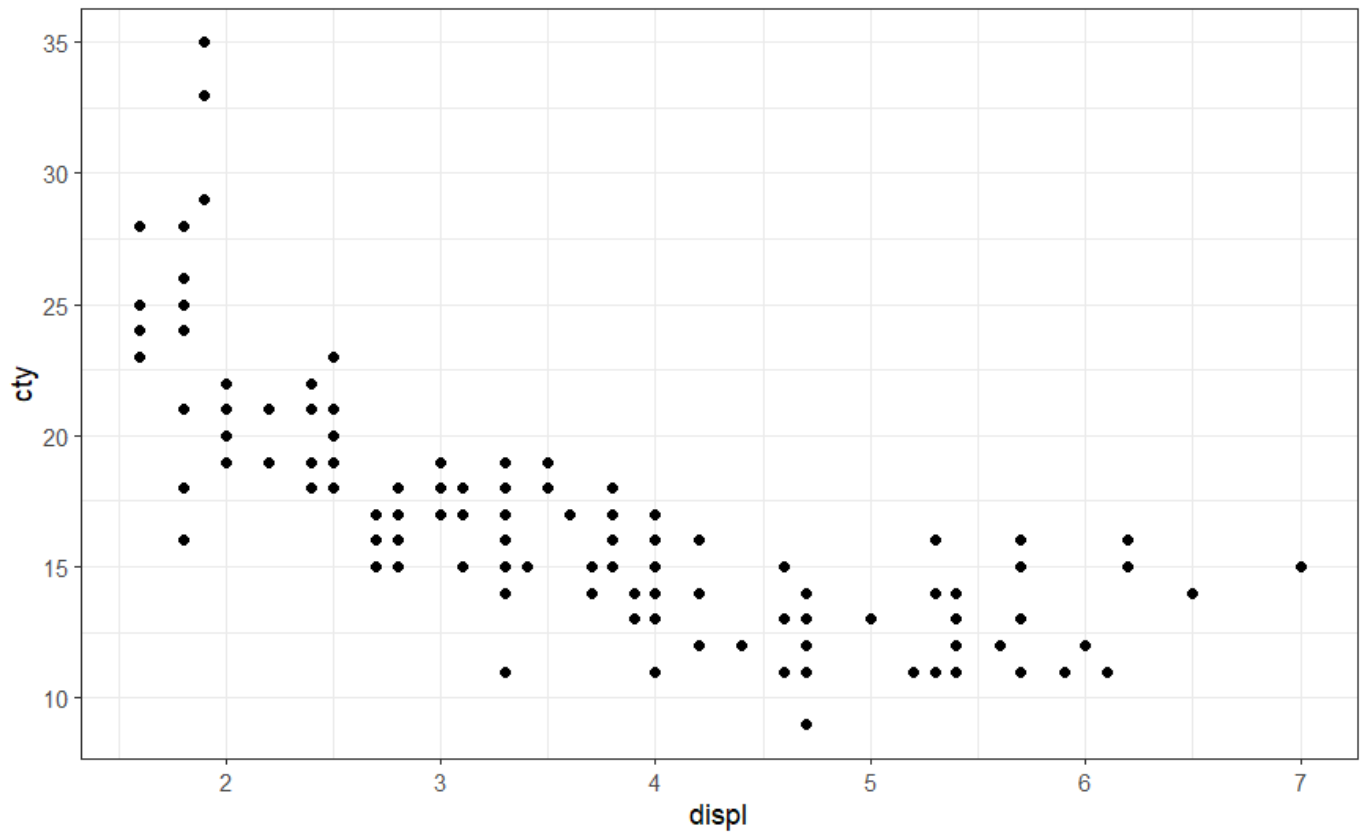
What does shape = class do? → It assigns a different point shape to each car class instead of color.

Which is better, shape = class or color = class? → Color is usually better here, because shapes are harder to distinguish when there are many points.

What about combining shapes and colors? Is this a good idea? themes? → It can help for accessibility (e.g., colorblind users) but can also make the chart look cluttered — so use it only when necessary.

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```
g <- ggplot(data = mpg,
mapping=aes(x=displ,
y=cty)) + theme_bw()
g+geom_point()
```

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```
g + geom_point(aes(color=as.factor(year)))
```

